

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

IV Semester of BSc (BT/MI/BC) Examination April-May 2018

Course code & Name: BE 405: Evolution and Behavior

Date: 08/05/2018 Day: Tuesday Time: 10:00 am To 10:20 am Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions. 10

1. Primates appeared during_____Era.
(i) Cenozoic (ii) Mesozoic
(iii) Paleozoic (iv) Protozoic
2. Hamilton's theory tries to explain _____.
(i) Multiple selection (ii) altruism
(iii) Evolutionary synthesis (iv) None of these
3. _____allows adaptation against low oxygen in Andes highlanders.
(i) Sexual selection (ii) Genetic drift
(iii) Positive selection (iv) None of these
4. According to molecular systematics the closest relative of the Dinosaur is_____.
(i) the Lion (ii) the Kangaroo
(iii) the Fish (iv) the Bird

5. Sense of smell helps avoid inbreeding.
- ☐ True ☐ False
6. Geological time scale can be divided into two large Eon segments.
- ☐ True ☐ False
7. Epigenetics can be linked to (quasi) Lamarckian evolution.
- ☐ True ☐ False
8. Natural selection acts on genetic diversity.
- ☐ True ☐ False
9. Habituation is an example of non-associative learning.
- ☐ True ☐ False
10. Genetic drift is linked to Sexual selection.
- ☐ True ☐ False

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IV Semester of BSc (BT/MI/BC) Examination April-May 2018

Course code & Name: BE 405: Evolution and Behavior

Date: 08/05/2018 Day: Tuesday Time: 10:21 am To 12:00 pm Maximum Marks: 25

Instructions:

1. Section I and II must be attempted in SINGLE ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION - I

Q - II Answer the following questions as directed (any two) 06

- (a) Discuss in brief about eusociality and altruisms.
- (b) Explain in brief about evolution by Natural Selection.
- (c) What is Genetic drift? How it is related to Bottlenecks and founder effects.

Q - III Answer the following questions as directed (any two) 04

- (a) Write a brief note on "Theory of Biogenesis"
- (b) Enlist the molecular tools in used in phylogeny.
- (c) Write a brief note on "Molecular Clocks"

SECTION - II

Q-IV Answer the following questions as directed (any three) 15

- (a) One popular, albeit untested, view suggests that Nagars are the descendants of many Greek and Macedonia soldiers who stayed in India after the Indian campaign of Alexander the Great (Sikandar)--which began in 326 BC. Can you test this view by applying the knowledge of molecular genetics? If so, what molecular technique is needed to test this view? Also discuss the theoretical basis of this molecular technique.
- (b) Discuss animal mating systems in animals.
- (c) Describe flexible and restrictive types of learning.
- (d) Discuss the differences between learning and memory. Please name all different types of memories.